



Learning to correct without breaking everything else

A frozen language model, an adaptive cap, and the consequences of rare errors

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35–40 minute colleague version

Active inference / Predictive coding / Heavy-tailed distributions

Active Inference in the Extremes · colleague rehearsal · results through 4 October 2026



Three questions, three different kinds of evidence

Active inference

What should the agent do next?

Beliefs, preferences and information seeking

Policy loop: proposed

Predictive coding

How should a correction learn?

Iterative inference of error signals

Learning comparisons: measured

Heavy-tailed distributions

How do consequences accumulate?

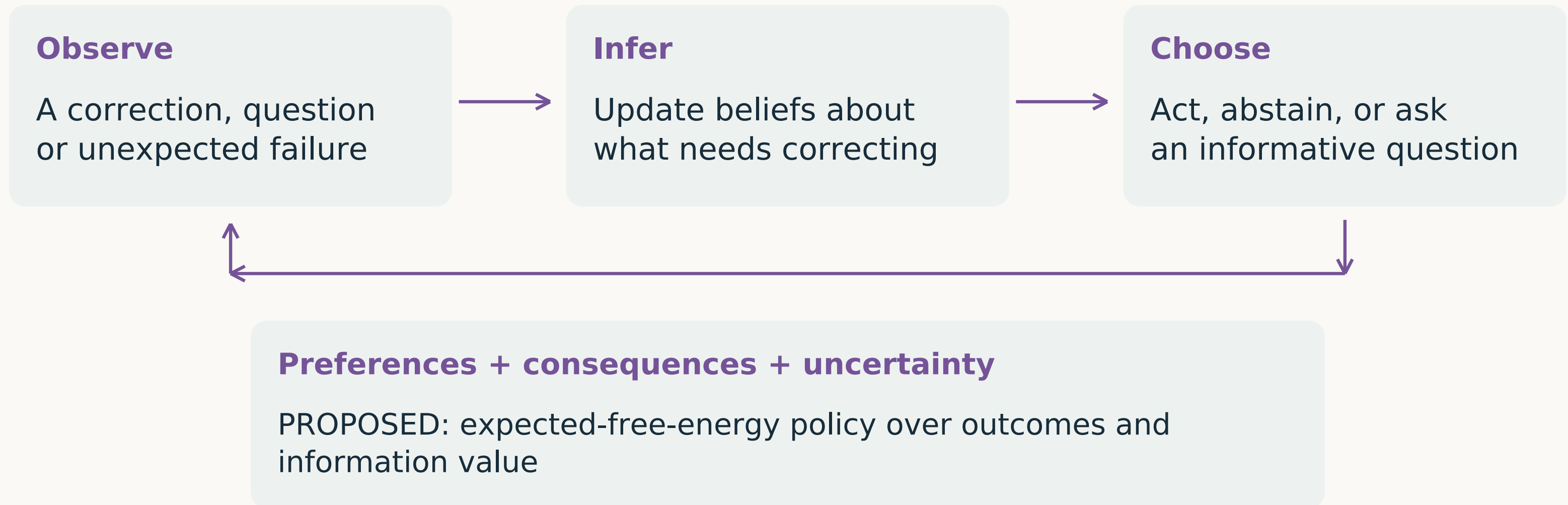
Frequency, severity and tail shape

Finite-range distributions: measured

The same testbed connects the questions; it does not make them interchangeable.



Active inference starts with a loop



Our investigators choose the audits today. An autonomous policy would have to choose them itself.



The cap changes an activation, not the frozen base weights

Environment

Supplied edits
New questions
Ordinary text



Adaptive cap

Fact memory
Learned reader + null gate
Bounded residual writes



Frozen transformer

Predict next token
Read internal features
Base weights fixed



A null decision leaves the base prediction in place.

An interface is implemented; a formal Markov blanket is not established.

The main system uses frozen GPT-2 small (124M parameters), with reads and writes at blocks 4, 8 and 12.



One edit has several tests to pass

1,000 edits for zsRE / CounterFact; 300 for MQuAKE. The first record has 999 / 299 subsequent edits.

Teach

Prompt + target answer

Immediate answer

ES: edit success

Return later

At the end of the edit stream

Original prompt / paraphrase

RET-ES / RET-GS

Look elsewhere

Unrelated and neighboring prompts

Ordinary-text probabilities

Preservation and harm

Acquiring an answer, remembering it, generalizing it and preserving other behavior are separate outcomes.



A recorded correction, after 1,000 edits

zsRE · realization 0 / order 100 · first stored item · 1,000-edit endpoint

Teach: What league was Sporting Canamy? → Tercera División de México

Paraphrase: What league did Sporting Canamy join with?

At the 1,000-edit endpoint	Original prompt	Paraphrase
Learned reader	“Tercera División de México”	“Tercera División de México”
Random reader	“Tercera División de México”	“CD Atlético Baleares”
Stable v0	[empty answer]	“forward”

All three acquired the target immediately. Cap-free original answer: empty (newline).

The learned reader retrieves this fact from a paraphrase; the other two conditions give different answers.



A memory can survive while its paraphrase fails

CounterFact · realization 0 / order 100 · 1,000 edits · deliberately counterfactual targets

PARAPHRASE SUCCESS

Teach: James Howell speaks

Target: Spanish (stored old: English)

Ask: Cataraqui is also the name of a municipal electoral district. The language used by James Howell is

Paraphrase output: “Spanish”

Cap fired: yes

PARAPHRASE FAILURE

Teach: Gol & Gincu The Series was created in the country of

Target: Philippines (stored old: Malaysia)

Ask: He completed a Ph.D. there in 1954. Gol & Gincu The Series originated in

Paraphrase output: “the late 1960s as a series of books by the late Dr. John Gol. The... [excerpt]”

Cap fired: no

Both taught targets survive on the original prompt. One paraphrase fires the learned cap; the other does not.



Preserving behavior can mean preserving a bad answer

zsRE · realization 0 / order 100 · saved endpoint probes at the 1,000-edit checkpoint

Teach: Where was Frits Poelman from? → New Zealand

Ask nearby: Where was George Pitt Morison from?

Condition	Neighboring answer	Preservation score
Base (no cap)	[empty answer]	Reference
Learned reader	[empty answer]	Preserved
Random reader	“New Zealand”	Changed
Stable v0	“Paris”	Changed

Locality example: “what is the name of fred flintstones wife” → “?” both with and without cap.

An unchanged answer establishes preservation on that prompt; it does not establish factual correctness.



What the main comparison holds fixed

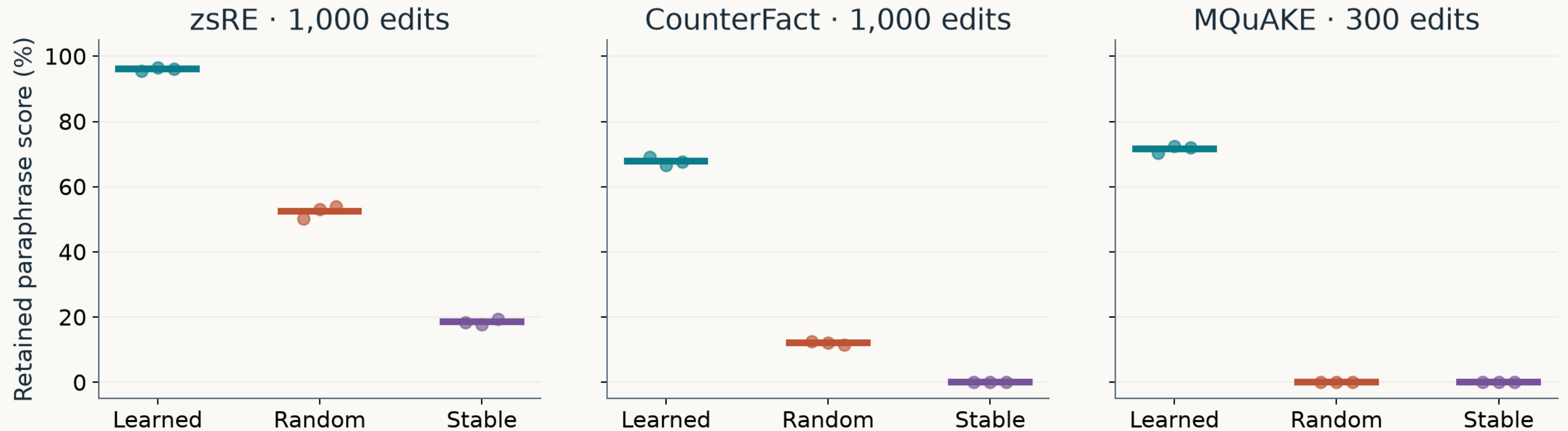
Learned reader · random reader + gate · stable v0 cap

Dataset	What is supplied	Endpoint
zsRE	Question → supplied factual answer	1,000 edits
CounterFact	Sentence stem → counterfactual target	1,000 edits
MQuAKE	Single-hop fact from MQuAKE data	300 edits; descriptive

The 270-cell snapshot includes further live-cap, matched-update and continued-base controls.

Three subject realizations × five orders. Reordering the same facts does not create five new populations.

The learned reader retains more paraphrases



Learned-reader means: 96.0% zsRE · 67.8% CounterFact · 71.6% MQuAKE (at 300 edits).



A changed probability can reveal harm before the answer changes

Same text prefix, same observed next token

ILLUSTRATIVE arithmetic

Base probability $p_0 = 0.10$

Cap probability $p_{\text{cap}} = 0.01$

Loss increase = $\log(10) \approx 2.30$ nats

ACTUAL main-study benchmark

45 / 45 learned-reader cells

exceed mean KL 0.001.

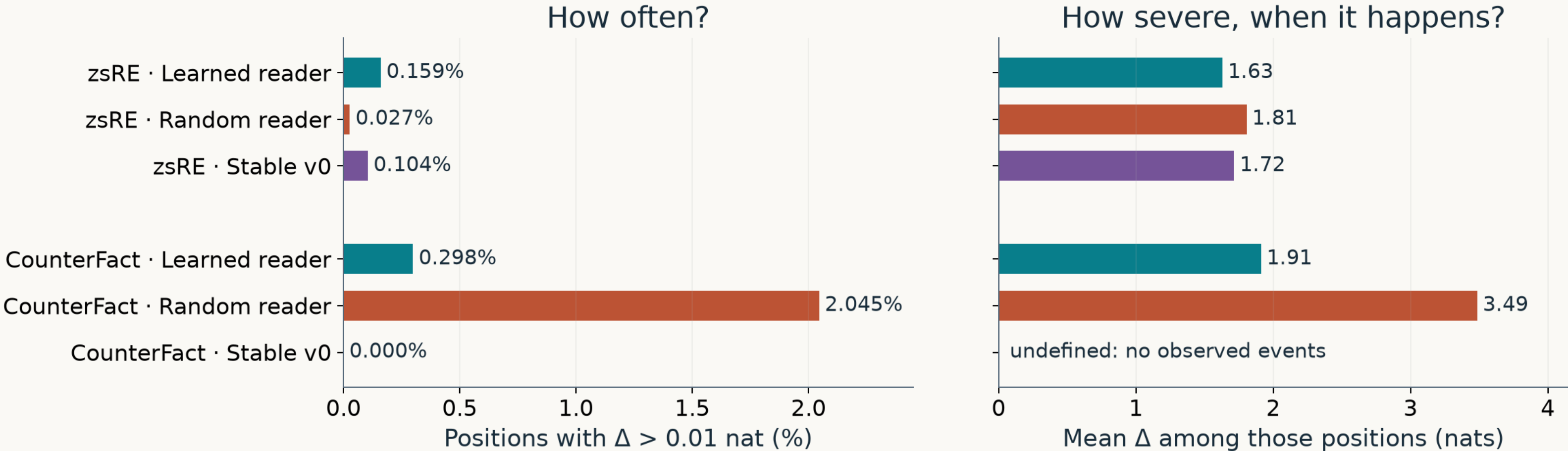
Their data-integrity checks pass.

Observed-token loss and whole-distribution KL answer different questions.

$\Delta = \log(p_0 / p_{\text{cap}})$ for the observed next token, at the same prefix. Positive Δ means worse prediction.



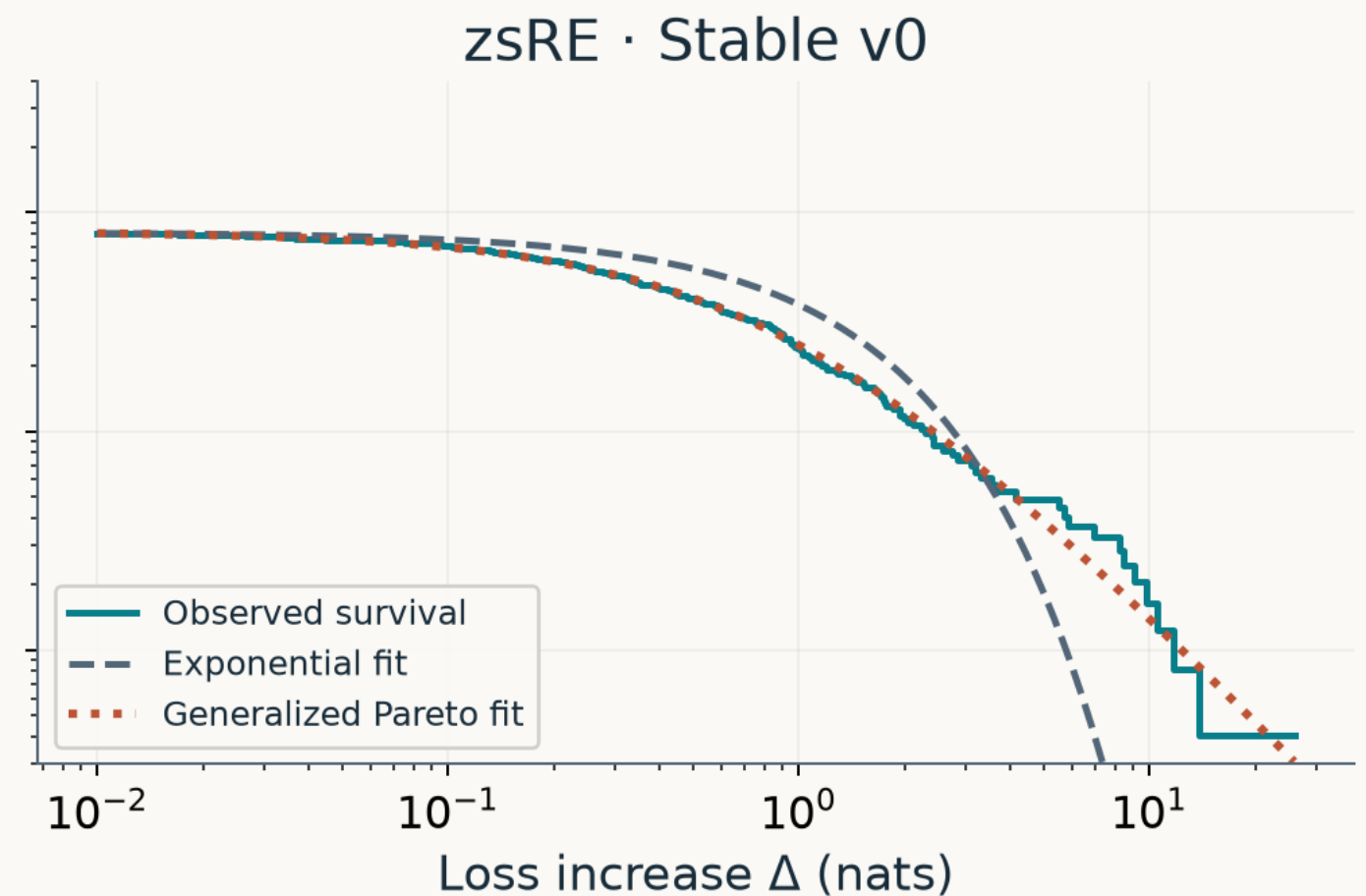
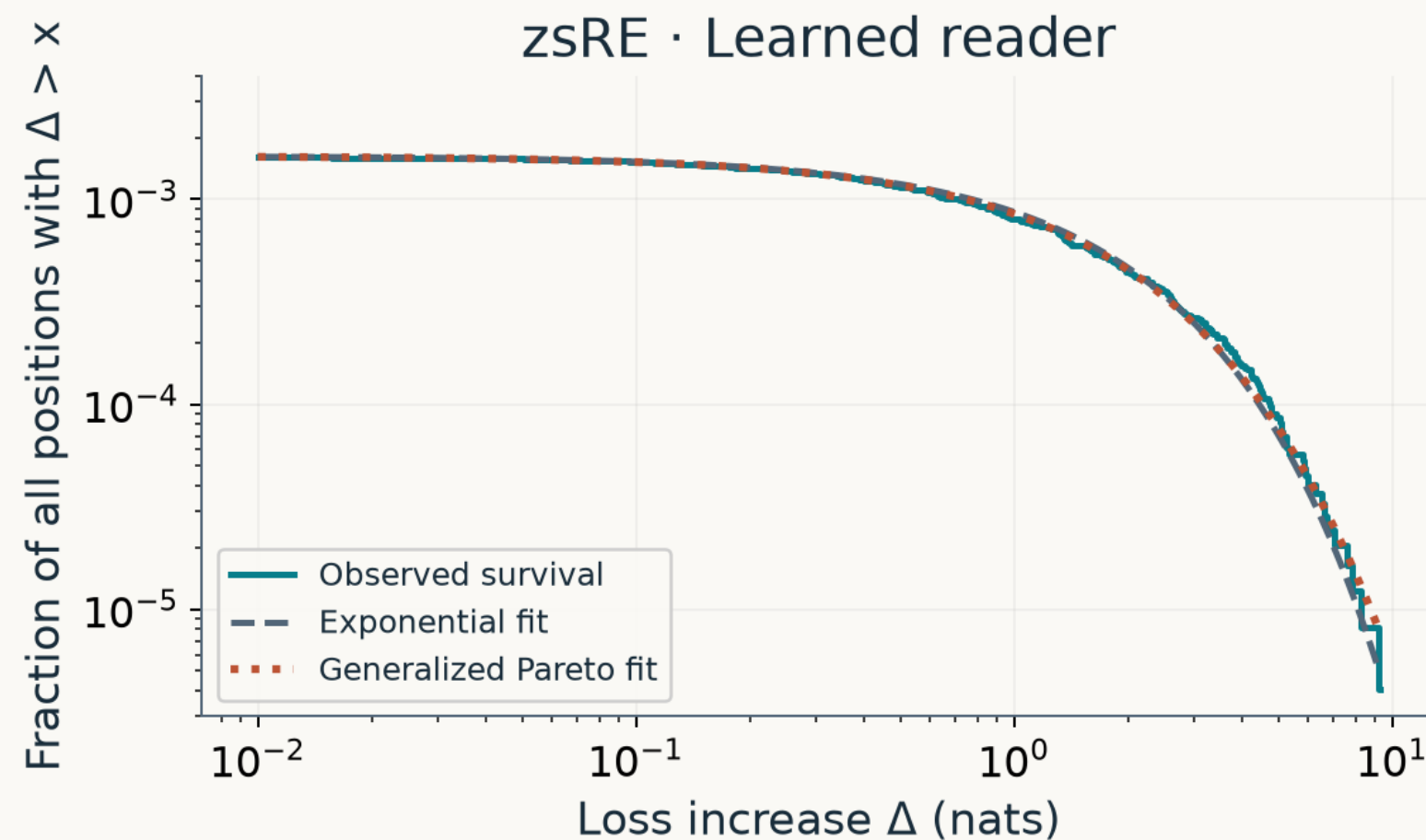
How often something goes wrong is not how badly it goes wrong



Same 245,237 ordinary-text positions per cell. Harm threshold: $\Delta > 0.01$ nat; gate firing is a different count.

Rare large errors motivate tail analysis; they do not prove a tail law

zsRE · realization 0 / order 100 · 1,000 edits · saved fits above 0.01 nat



Learned v5: little held-out gain over an exponential. Stable v0 on zsRE: a more pronounced finite-range tail.

A simple mixture gives a real per-token bound

Frozen base

37% base distribution

$$\rho = e^{-1} \approx 0.368$$

Corrected model

63% cap distribution

$$1 - \rho \approx 0.632$$

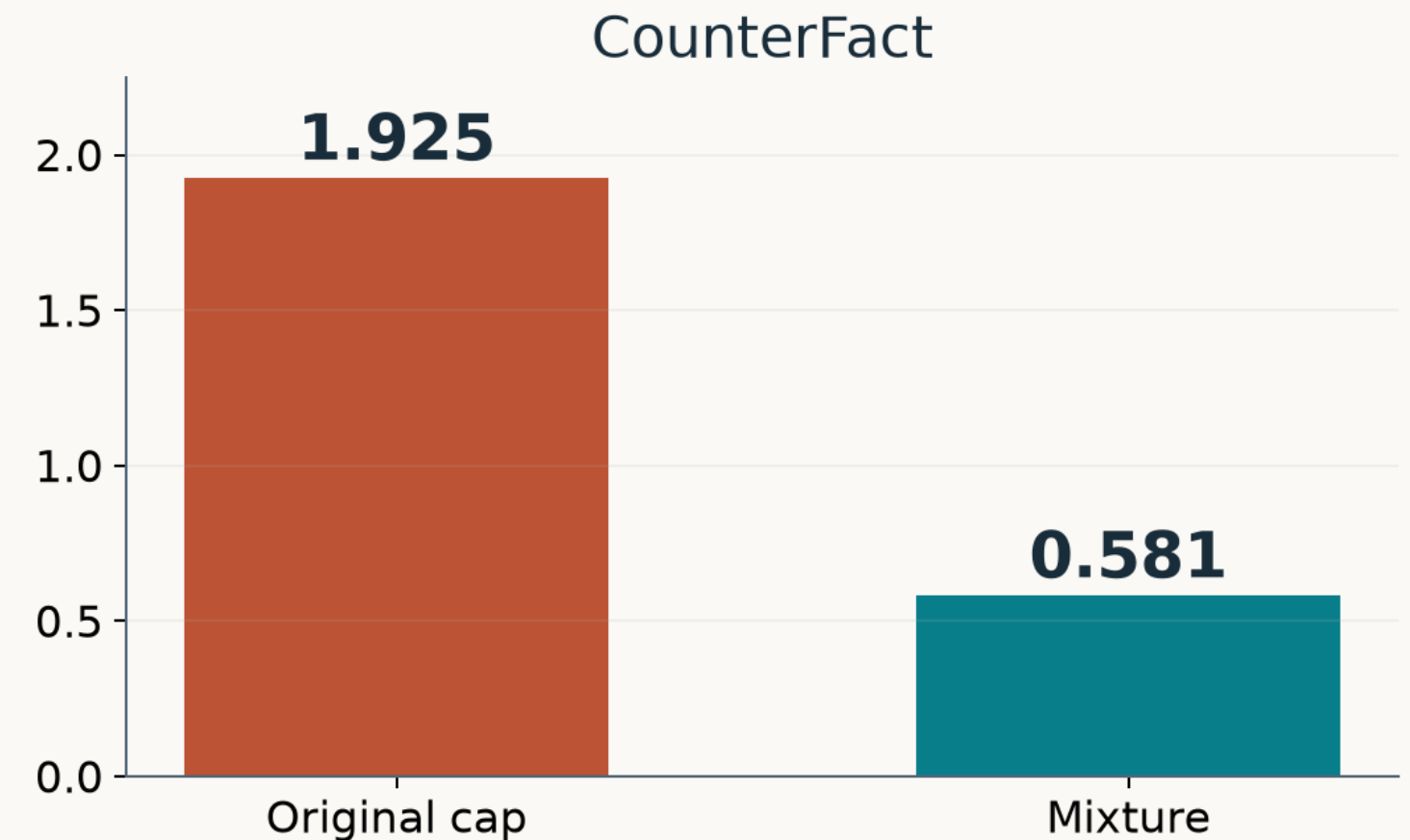
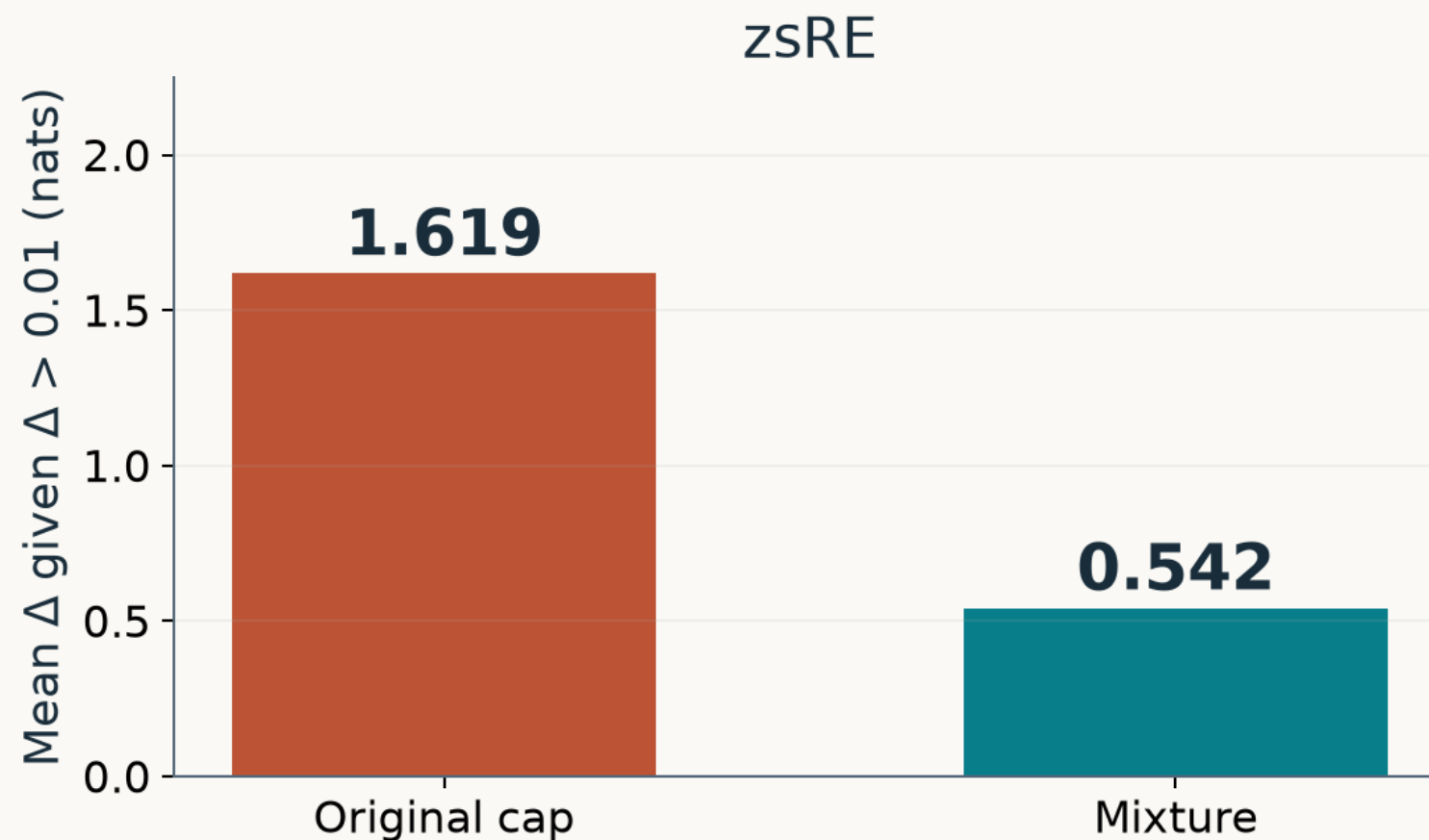
Probability mixture

$$q = \rho p_0 + (1 - \rho) p_{\text{cap}}$$

Every token retains a floor of ρp_0 .

At a shared prefix, $q \geq \rho p_0$. Therefore $\log(p_0/q) \leq -\log \rho = 1$ nat when $\rho = e^{-1}$.

The mixture reduced severity while largely retaining the edits



Ten exposed memories pass the declared rule. Conditional severity falls to roughly one-third; frequency changes little.



Predictive coding: infer errors before choosing an update

Adjoint credit · SE-A

Differentiate the task loss
Take a negative normalized direction
Apply the bounded memory update

Error inference · SE-E

Set temporary errors $e = 0$
Minimize $E(e) = \frac{1}{2}\sum \|e\|^2 + \text{task loss}$
Normalize inferred site errors

Same target, fixed existing writes, same bounded update

8 settling steps at $\eta = 0.1$; 9 forward + 9 reverse evaluations including the terminal diagnostic.

Both paths feed the same bounded acquisition update. Our error inference uses JAX differentiation.



A repaired energy and a useful one-step sanity check

Corrected objective

$$E(e) = \frac{1}{2} \sum \|e\|^2 + \text{task loss}$$

Existing writes stay fixed during inference.

Old penalty used $e + w$.

First step from $e = 0$

$$e_1 = -\eta \times \text{adjoint}$$

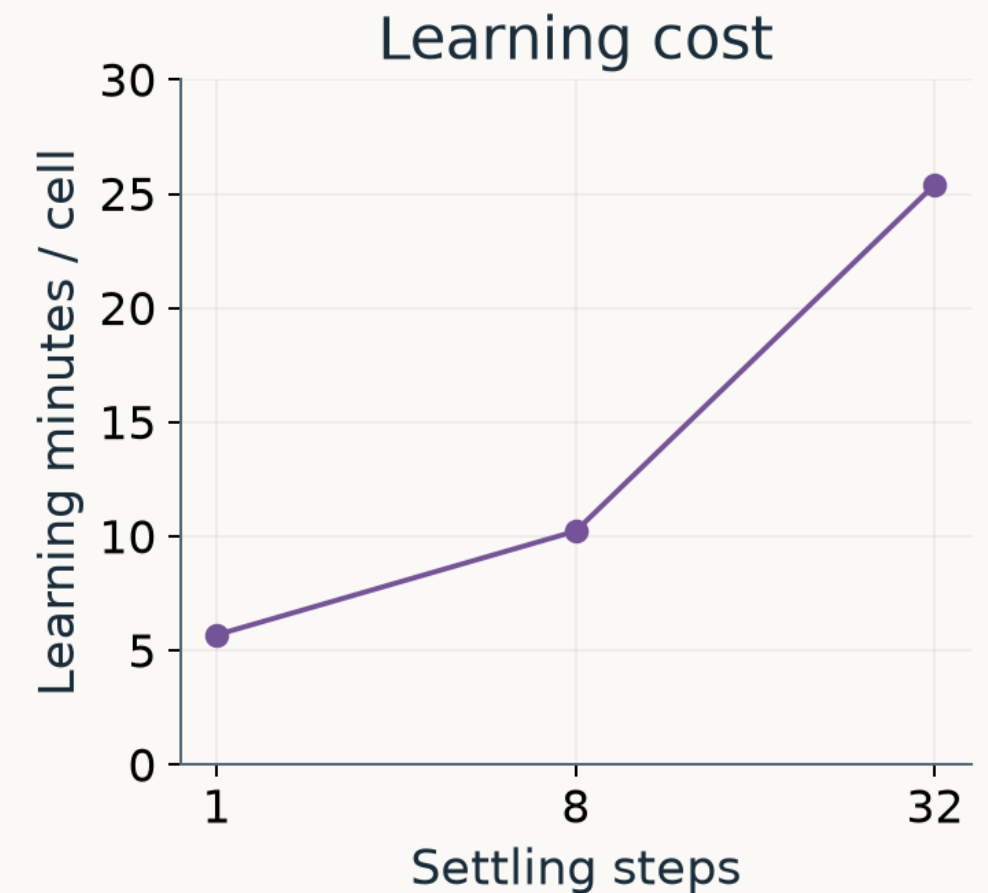
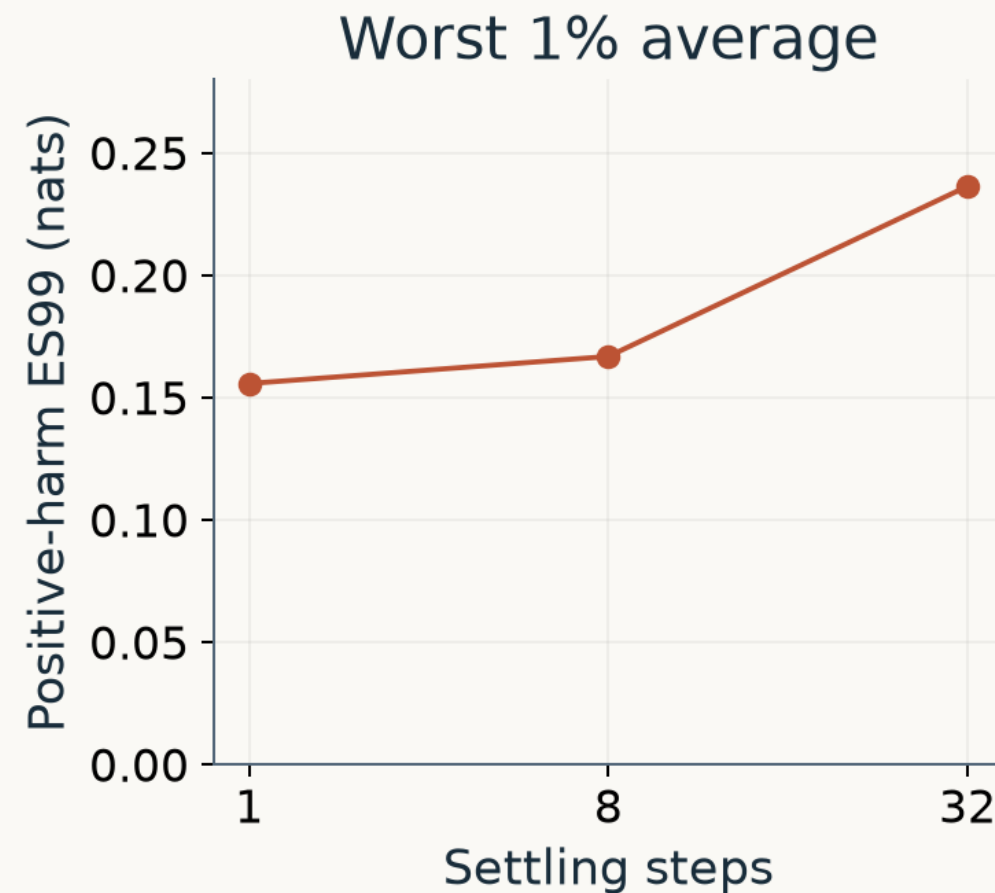
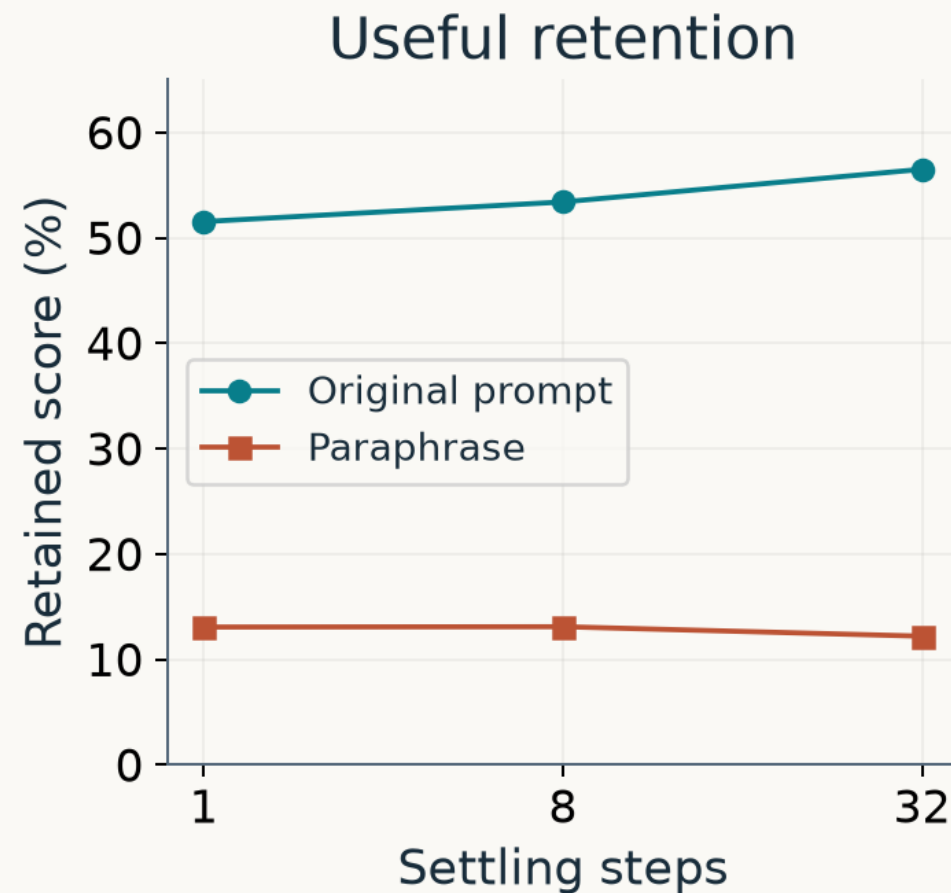
Normalization removes $\eta > 0$.

Deeper settling can change direction.

The long Stage-4 feedforward run did not use the defective PC credit rule. The PC results here use the correction.



More settling bought memory retention, with costs



Legacy 50M base, v0 live cap; zsRE at 1,000 edits. Three realizations, order 100; exposed populations.



Changing credit on the working reader gave small, mixed effects

Same reader; only the acquisition credit changes

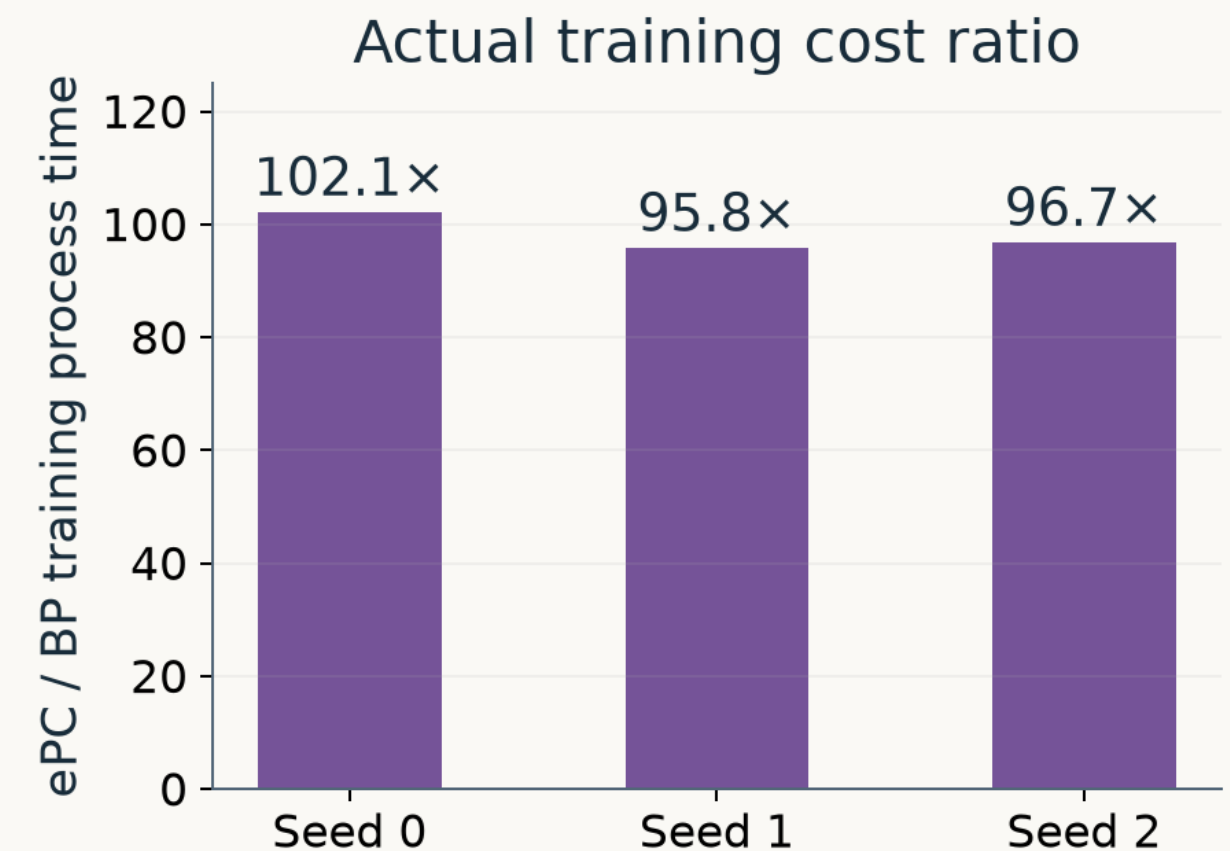
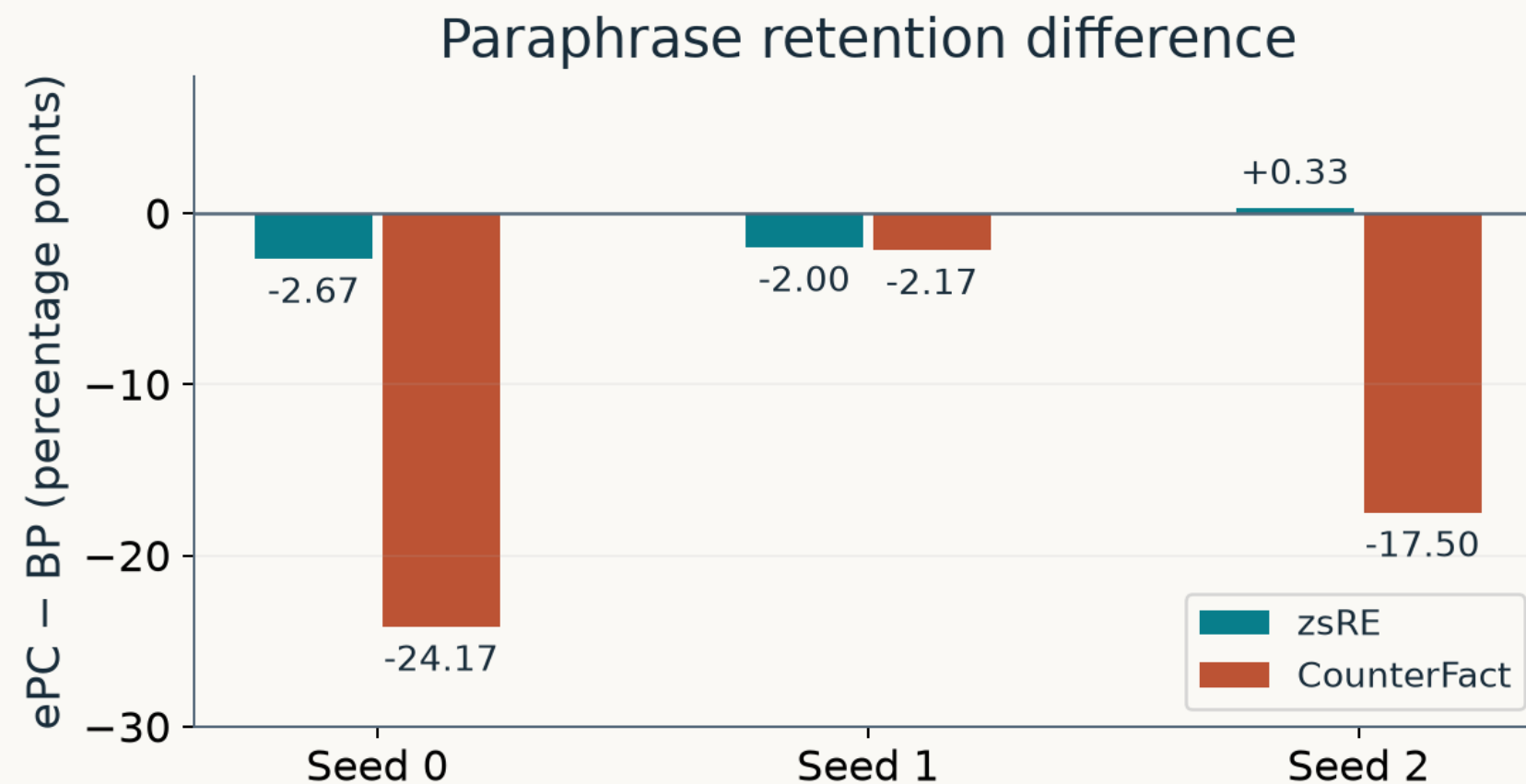
SE-E minus adjoint	zsRE	CounterFact
Paraphrase retention	0.00 percentage points	−0.33 percentage points
Positive-harm ES99	−0.01397 nats	+0.01123 nats

Stream totals: adjoint 604.6 s → error inference 776.7 s. Separate harm readout: 5,004 s.

Fixed BP-trained v5 reader; fresh memories; 300 edits. Four cells, one exposed subject realization per dataset.



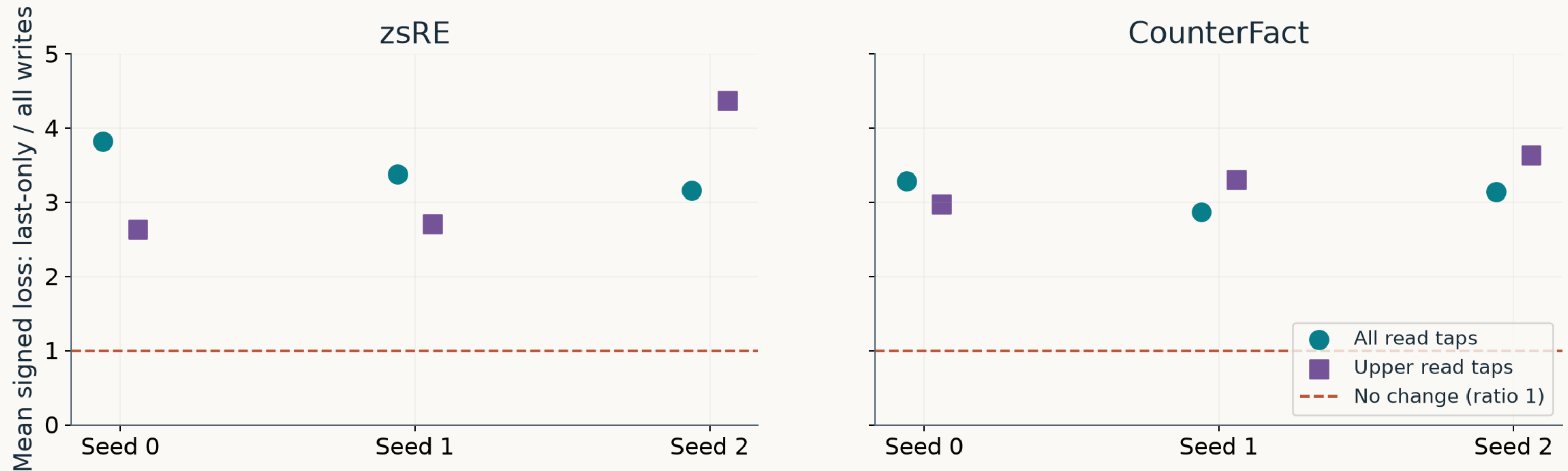
Training the reader with PC was much more expensive



Three paired training seeds, same exposed subjects; acquisition is adjoint in BOTH arms. Completed: 12/12 evaluations.



Restricting the cap to upper layers did not remove the harm



Last-site-only writes increased mean loss 2.64-4.37× in all 12 write pairs, with little paraphrase change.



The next agent could choose which failure to investigate

Available evidence

Retained corrections
Wrong-memory retrieval
Rare ordinary-text losses



PROPOSED policy

Beliefs + preferences
Action / observation
model
Information value and
cost



Choose one audit

Original prompt?
New paraphrase?
Nearby unedited fact?

Coupled-objective collaboration: first specify the probability model

Modeled variables · constraints · weighting / escort distribution · gradients · explicit baseline

Proposed experiment: policy-selected audits versus fixed audits, with the same audit and compute budget.



The experimental portfolio is closed; the story is still being refined

Completed evidence

270-cell main snapshot
PC credit + reader studies
Mixture, tails and upper layers

Practical limits

Small models; limited populations
Some controls closed by resource rule
Main snapshot: 392.42 process-hours

Before Binghamton

Review fixed evidence
Choose the essential story
Rehearse the 15-minute version

Experiments completed or explicitly closed by 4 October. Review freeze: 9 October, 17:00 EDT. Talk: 15 October.



Three takeaways to carry into the conference

Active inference

A testbed for selective correction

Autonomous audit policy remains open

Predictive coding

Measured learning-rule trade-offs

No general advantage in this recipe

Extremes

Means miss consequential structure

Mixture bounds per-token loss

The scientific contribution is a measured trade-off among useful correction, unintended harm and learning cost.



Help choose the fifteen-minute story

Keep

Which example or graph stayed with you?

What would you tell someone tomorrow?

Clarify

Where did the explanation lose you?

Which claim needs more evidence?

Cut or move

What belongs in Q&A?

Which detail can the short talk omit?

Please give slide numbers: one to keep, one to clarify, and one to move to questions.



MQuAKE: a single-hop editing example

Realization 0 / order 100 · 300 edits · deliberately counterfactual targets; no multi-hop test

PARAPHRASE SUCCESS

Teach: The type of music that Marion Brown plays is

Target: West Coast hip hop (stored old: jazz)

Ask: What type of music does Marion Brown play?

Paraphrase output: “West Coast hip hop”

Cap fired: yes

PARAPHRASE FAILURE

Teach: Bettino Ricasoli is affiliated with the religion of

Target: Methodism (stored old: Catholic Church)

Ask: Which religion is Bettino Ricasoli affiliated with?

Paraphrase output: [empty answer]

Cap fired: yes

Same realization 0 / order 100; 300-edit endpoint. A fired cap can still fail to produce the target.



The risk summaries answer different questions

Quantity	Definition / question
Signed mean Δ	Average $\log(p_o/p_{cap})$; benefits remain negative
Frequency / severity	$P(\Delta > u) / E[\Delta \mid \Delta > u]$, here $u = 0.01$ nat
ES99+	Fractional mean of worst 1% of $\max(\Delta, 0)$, all positions
Mean KL	Average $KL(p_o \parallel p_{cap})$ over next-token distributions
Maximum	Largest observed Δ ; not an unseen-event guarantee

The token-position inventory is paired and dependent. None of these quantities turns tokens into subject replicates.



What the extra PC controls do and do not settle

Random direction

3 immediate successes / 993 edits

One attempted random stream

Resource stop; no final retention claim

Additional-update adjoint

12 cells / 6 complete pairs

About 53–58% of offered operations used

Direction / effective compute unresolved

No equal-realized-compute superiority claim follows from an allowance that the comparator did not spend.



A fourth subject draw supports the learned-random contrast

Fourth realization	Learned RET-GS	Random RET-GS
zsRE	96.9%	51.5%
CounterFact	68.15%	10.6%

Four-draw contrast: zsRE +44.1 pp; CounterFact +56.2 pp. No new primary decision rule.

Supplemental sensitivity analysis. Deferred v0 cells stay absent; the original confirmatory classifier is unchanged.



Source map and the submitted programme

Core evidence

R1_stage4_report.md

HT-17_report.md

Capstan's
support-information
examples

Learning experiments

PC-v0 / PC-v1 reports

PC-reader_report.md

AW-L_report.md +
AW-B_report.md

Research programme

July submitted abstract

Presentation brief, 26
September

Coupled collaboration
proposal

Full source paths, example identities and hashes accompany the deck. No new experiment was run to prepare it.